MILITARY SPECIFICATION

FILTER ELEMENT, FLUID PRESSURE FFU-5/E

1. SCOPE

1.1 This specification covers one type of fluid pressure filter element, designated FFU-5/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	Box, Setup
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed

FSC 4330

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications . The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

D 256	Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials
D 638	Method of Test for Tensile Properties of Plastics
D 790	Standard Method of Test for Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Materials

3.3.1 Metallic parts. All metallic parts of the filter element shall be fabricated from an aluminum alloy suitable for the application.

3.3.2 Fungusproof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.3.3 Petroleum products. The materials used in the construction of the filter element shall neither affect nor be affected by petroleum products.

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3.3.4 Plastic parts. If used in the construction of the elements, molded plastic parts shall have the following physical properties:

a. Tensile strength minimum . . . . . 4.500 psi

b. Flexural strength minimum . . . . . 7.500 psi

c. IZOD impact minimum 1.5 pounds per inch of notch.

3.4 Design and construction. The filter element shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.4.1 Functional design. The filter element shall be designed for use in the Bowser, Incorporated, Bowser-Briggs Filtration Division series 842D filter-separators. The element shall be as shown on figure 1. The element shall be provided with an aluminum-alloy center tube. The element base shall be provided with aluminum alloy threads of the size shown on figure 1. If the threaded section is a metallic insert, it shall have a minimum outside diameter of 2-1/4 inches. The threads shall be coated with a material to prevent the element from loosening in service after installation. The coating shall not prevent the element from being removed. A 3/8-16 UNC-2B stainless-steel nut, flat stainless-steel washer, and Buna N gasket of suitable size shall be provided to attach the canister to the element and prevent leakage around the stud.

3.4.1.1 Compatible fuels. The filter element shall be so designed that it can be used with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.4.1.2 Gasket. A gasket shall be furnished in each element. The gasket shall have a 1-1/2-inch inside diameter by 2-inch outside diameter and shall be 1/8 inch thick. The material for the gasket shall be Buns N or other material compatible with fuels conforming to MIL-T-5624.

3.4.2 Separator canister assembly. When specified (see 6.2), a separator canister shall be furnished with each element. The canister shall be a semipermanent separator element normally not needing replacement, and made entirely from metallic parts except that the end cap may be made of polyester resin as specified herein, and adhesives may be used, if required. It shall be of rigid construction and shall be designed to withstand damage during normal handling and installation. The canister dimensions shall be as shown on figure 2. The canister shall be constructed of two concentric layers of perforated metal and wire mesh. The outer layer shall be a covering made from aluminum alloy metal having a minimum thickness of 14 gage. The perforations shall be 3/16-inch-diameter holes on 5/16-inch centers minimum or 1/4-inch-diameter holes

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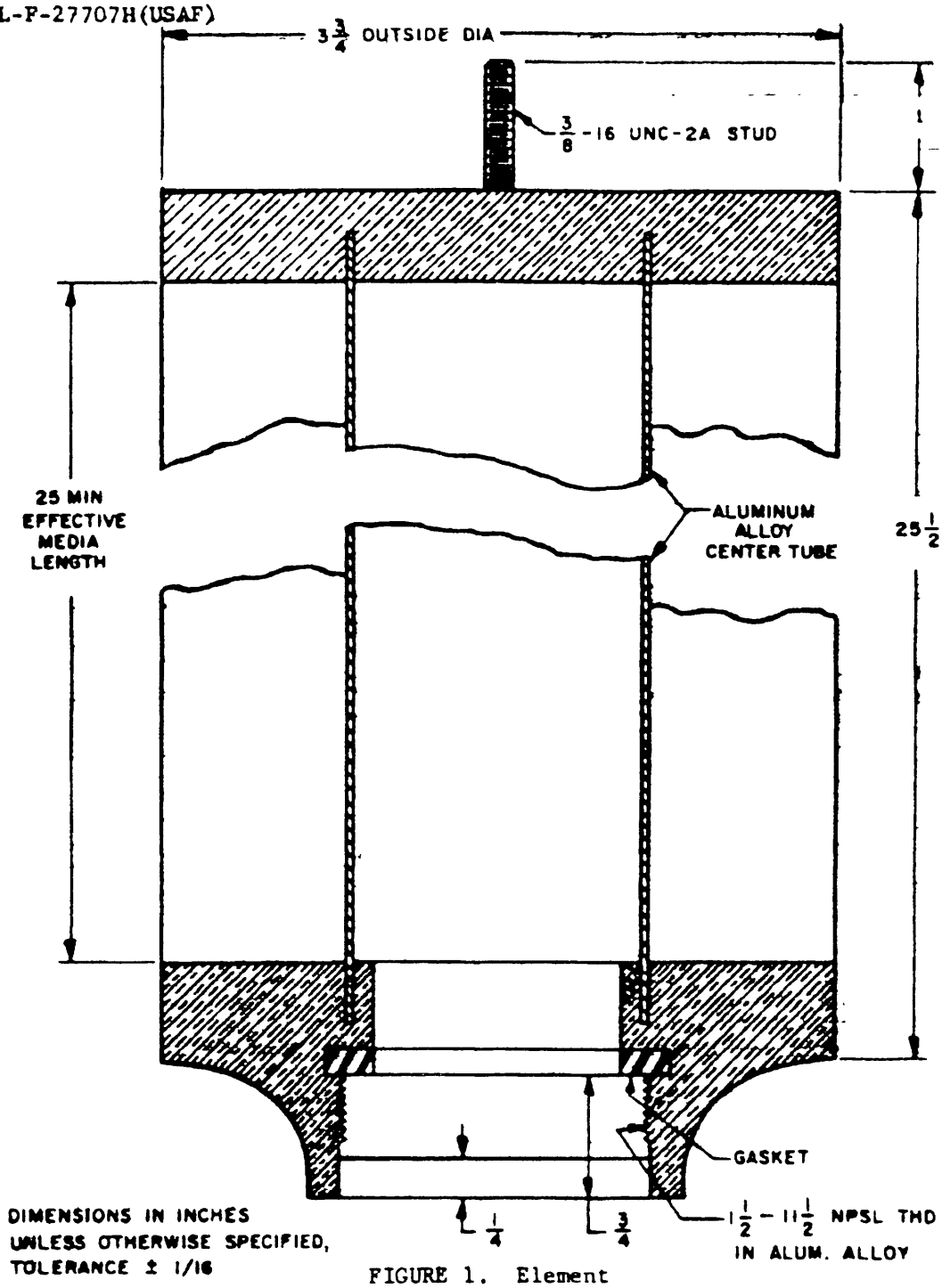


FIGURE 1. Element

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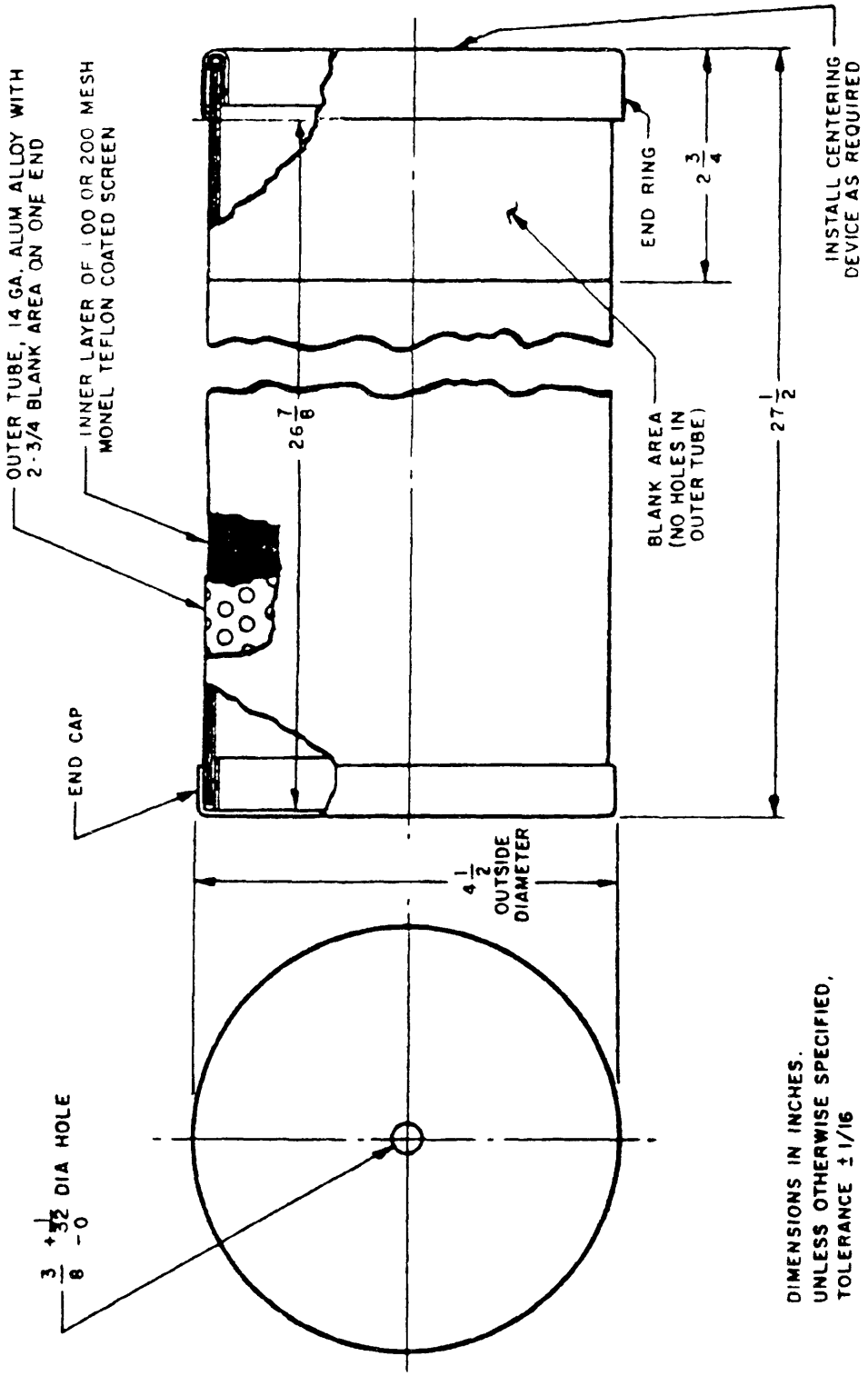


FIGURE 2. Separator Canister

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on 9/16-inch centers maximum. The inner layer of material shall be a hydrophobic medium of 100- or 200-mesh monel wire screen, 0.0021 inch in diameter, coated on both sides with teflon. The canister shall have a hole in the top section for fastening to the element shown on figure 1. The. canister shall be provided with a centering and securing device located in the bottom section to prevent movement and end play during service. The securing device shall in no way cause damage to the element during installation or operation.

3.5 Performance. When 24 elements and canisters are installed in a Bowser, Incorporated, Bowser-Briggs Filtration Division model 842D-600 filter separator, they shall be capable of meeting the performance requirements of MIL-F-8901 at a flow rate of 600 gpm.

3.6 Identification of product. The following information shall be marked on each filter element and each separator canister in accordance with MIL-STD-130. If the item is marked on any sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's part No. _____
Manufacturer's name or trademark _____
Date of manufacture _____

3.6.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

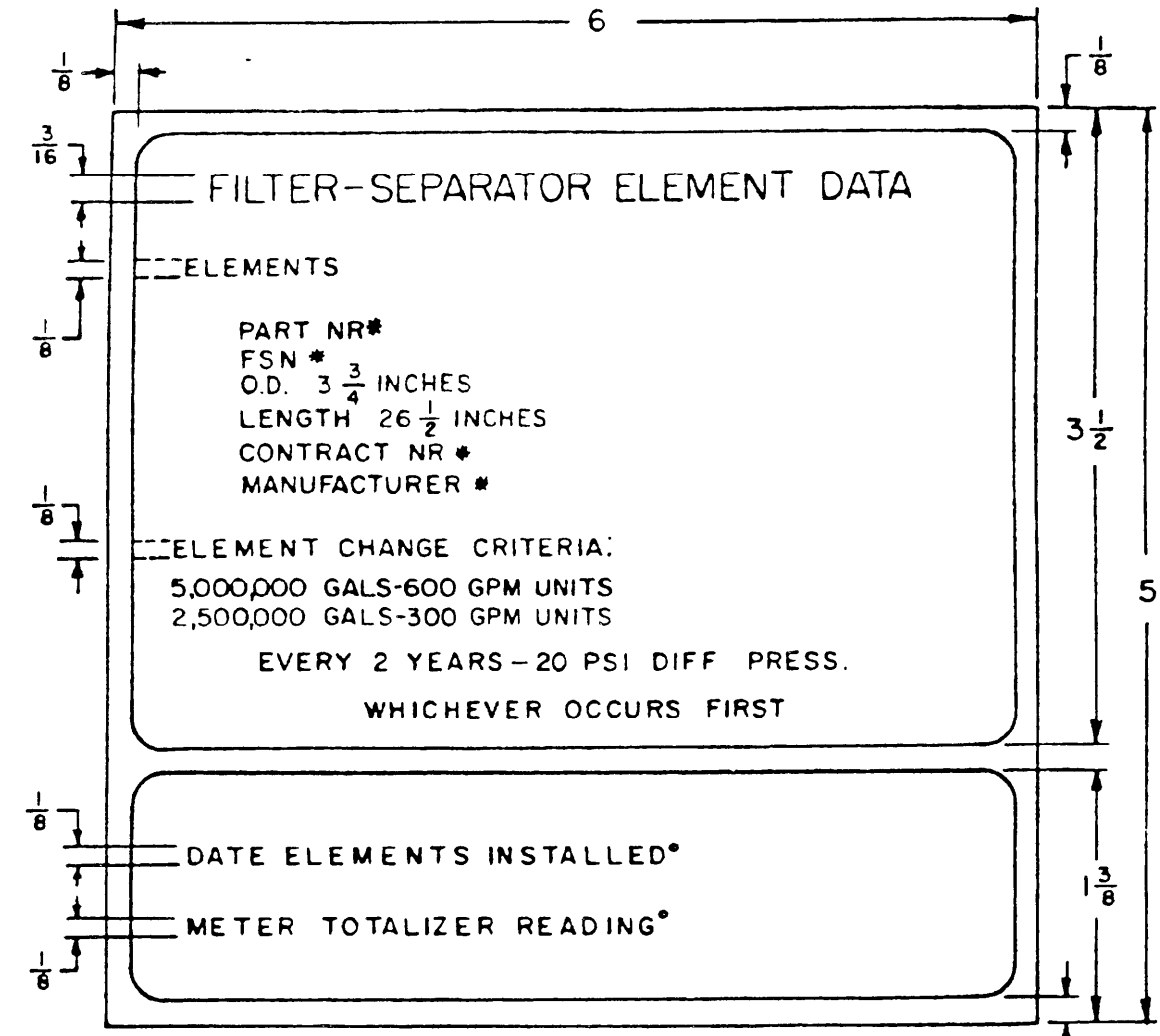
3.6.2 Nameplate. A nameplate fabricated from materials as specified in MIL-P-19834, type I, shall be supplied one per each 12 elements shipped. The nameplate is intended to be attached by Government personnel to the pressure vessel to facilitate maintenance of the fueling system. Attachment instructions shall be furnished, attached to the plate. The namplate shall conform to figure 3.

3.7 Workmanship. The filter element, including all parts, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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NOTES:

1. ALL LETTERING SHALL BE 3/32-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCE ± 1/32

FIGURE 3. Nameplate

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4.2 Test condition. During all applicable full-scale testing, 24 filter elements and 24 separator canisters shall be mounted in a Bowser, Incorporated, Bowser-Briggs Filtration Division model 842D-600 filter separator.

4.3 Preproduction testing (see 6.2)

4.3.1 Test samples. Unless otherwise specified (see 6.2), the preproduction test samples shall consist of 100 elements and 24 separator canisters representative of a production lot (see 6.3.1). The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. Samples shall be prepared as follows:

4.3.1.1 Data to accompany test samples. The preproduction test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment

b. Engineering data in the form of assembly drawings (two sets).

4.3.1.2 Test sample and data for procuring activity. When tests are conducted at a site other than the laboratory of the procuring activity, the following shall be furnished that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.

b. A sample element and separator canister removed from the filter-separator which was subjected to the preproduction tests

c. Color photographs of the sample element

d. A new filter element

e. A new separator canister (if applicable).

4.3.2 Preproduction tests. The preproduction tests shall consist of all tests specified herein and in MIL-F-8901, except as follows:

4.3.2.1 Separation tests. Water separation tests at 115 percent of rated capacity and at 10 percent by volume are not applicable. The separation tests shall be performed at 600 gpm.

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4.3.2.2 Inhibited fuel test. During the inhibited fuel test, a single-pass transfer system is required. Single element tests devised to represent full scale testing shall consist of the element shown on figure 1 installed in a canister as shown on figure 2. Single element testing shall be performed at a flow rate not less than 25 gpm.

4.3.2.3 Coalescer test. The filter element shall be subjected to the test of MIL-F-8901 entitled Coalescer Element.

4.3.2.4 Flow rate. During the Full-scale test conditions, the flow rate capacity for the filter element shall be 600 gpm.

4.3.2.5 Disassembly inspection. Upon completion of the Life test specified in MIL-F-8901, the filter-separator housing shall be disassembled and inspected to determine the condition of the filter elements. Defects in the filter element such as swelling of the element, damaged elements, dinged gaskets, et cetera, shall be recorded. Excessive swelling of or damage to the element or other parts shall be cause for rejection.

4.3.2.6 Plastic material test. Sections of plastic material used in the construction of end caps and elements or especially prepared test specimens shall be tested as follows to verify plastic material compliance. This test data developed shall be included in the test report:

a. Tensile strength - ASTM D 638, type I

b. Flexural strength - ASTM D 790

c. IZOD impact strength (notched) - ASTM D 256, method A.

4.4 Acceptance tests. The acceptance tests shall consist of the following:

8. Individual tests . . . . . See 4.4.1

b. Samping test . . . . . See 4.4.2.

4.4.1 Individual tests

4.4.1.1 Examination of product. Each element and canister shall be examined as required by the Examination of product paragraph of MIL-F-8901, and to determine that it conforms to the elements and canisters used in the preproduction tests.

4.4.1.2 Separator canister inspection. Each separator canister shall be subjected to an inspection to determine if all joints are properly sealed and the wire mesh is properly attached to insure integrity. The canister shall be

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mounted in a fixture having the capability of rotating the canister 360 degrees in a level plane. At least 1/2 inch of water shall be placed inside the canister. The canister shall be rotated 360 degrees and the element inspected for leakage. If leaks occur, it shall indicate defects in the hydrophobic medium (teflon screen).

4.4.2 Sampling plan and test. Samples of the filter element shall be selected at random from each lot in accordance with MIL-STD-105 and subjected to the test specified in 4.3.2.3. An AQL of 1.5 percent defective shall be established. The inspection level shall be level I. This test shall be considered as being destructive.

4.5 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2)

5.1.1 Level A. One filter element with adapting hardware shall be packaged in accordance with method III of MIL-P-116 in a container conforming to PPP-B-566, PPP-B-676, or PPP-B-636.

5.1.2 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing (see 6.2

5.2.1 Level A. Filter elements shall be packed in overseas-type containers conforming to PPP-B-576, PPP-B-636, PPP-B-640, PPP-B-585, or PPP-B-591. Containers shall be of minimum weight and cube, uniform size and shape, and shall contain like quantities, when practical. Closure of the containers shall be in accordance with the applicable specification and appendix thereto. Containers in accordance with PPP-B-636 shall be sealed at all joints and seams with tape conforming to PPP-T-60. The nameplate (see 3.6.2) shall be packaged in the same carton with each 12 elements or fraction thereof shipped.

5.2.2 Level B. The filter element shall be packed as specified in 5.2.1, except containers shall be of the domestic type and closure shall be in accordance with the requirements for domestic containers.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will insure safe delivery at the lowest transportation and packing charges. Containers

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shall meet consolidated freight rules and regulations or other rules and regulations applicable to the mode of transportation. The nameplate (see 3.6.2) shall be packaged in the same carton with each 12 elements or fraction thereof shipped.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see 6.2) all shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

FILTER ELEMENT, FLUID, PRESSURE FFU-5/E

6. NOTES

6.1 Intended use. The FFU-5/E filter element is intended for use as a replacement item in ground aviation liquid fuel filtering and servicing systems. This filter element and canister are intended to replace Bowser-Briggs Division part No. 47B37 and A2653 elements and canisters in the Bowser, Incorporated, Bowser-Briggs Division model 842D-300 and 842D-600 filter-separators.

6.1.1 Canisters. It is intended that the canisters will replace the currently used separator bags in the Bowser, Incorporated, Bowser-Briggs Filtration Division model 842D-300 and 842D-600 filter-separators, The canisters are not intended to be replaced each time the element is replaced. Thus the canisters are intended to be furnished on a one-time basis for the above filter-separators.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When a separator canister is to be furnished (see 3.4.2)

c. Location and conditions for preproduction testing (see 4.3)

d. When the test samples are to be other than as specified in 4.3.1

e. Applicable levels of presentation, packaging, and packing (see 5.1 and 5.2)

f. Special shipment marking (see 5.3)

6.3 Definition. For the purpose of this specification, the following definition will apply:

6.3.1 Lot. All filter elements of the same part number manufactured in one production run and submitted for acceptance at the same time shall be considered a lot for acceptance purposes.

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6.4 Contract administration

6.4.1 The contracting organization should require that a prospective contractor shall have or can contract for suitable test system prior to the award of any contracts for filtration equipment. A test system is the piping, test equipment apparatus, arrangement of test components, laboratory equipment, laboratory supplies, and techniques used during the evaluation of fuel filtration equipment.

6.4.2 The contracting organization should provide for Government inspection during the preproduction buildup and test in addition to providing Government inspection after production is released.

6.4.3 The contracting organization should, prior to issuance of a procurement bid or proposal (IFB/RFP), contact the engineering activity having technical responsibility for the specification to confirm that the latest edition of the specification is being cited.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewers:
Air Force - 84
DSA-CS

Preparing activity:
Air Force - 11

Project No. 4330-F109

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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